

MyHealthReport

Technical Module Architecture

A modular backend design: UI, medical logic, laboratory data,
risk algorithms and reporting evolving independently

24 August 2026

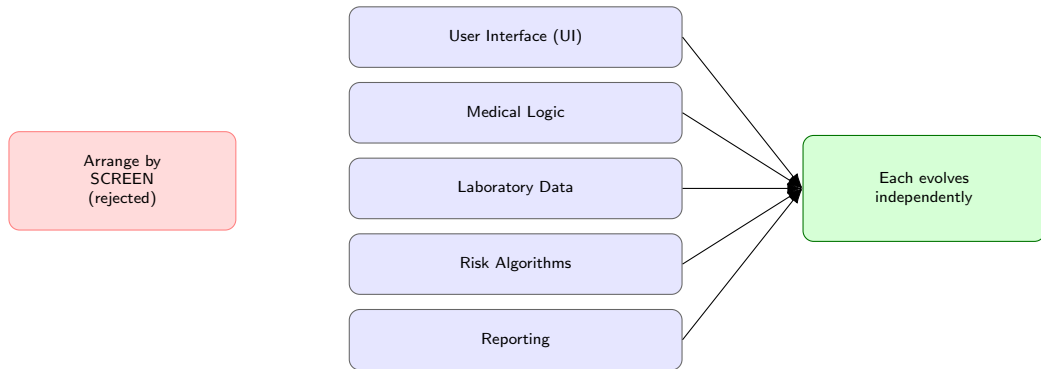
Agenda

- 1 Guiding Principle
- 2 Master Technical Architecture
- 3 Cluster 1 – Identity & Profile (M1–M4)
- 4 Cluster 2 – Data Ingestion & Standardisation (M5–M9)
- 5 Cluster 3 – Medical Intelligence Core (M10–M13)
- 6 Cluster 4 – Interpretation & Reporting (M14–M19)
- 7 Cluster 5 – Engagement (M20–M21)
- 8 Cluster 6 – Platform Governance (M22–M26)
- 9 Architecture Principles
- 10 Team Division

Guiding Principle

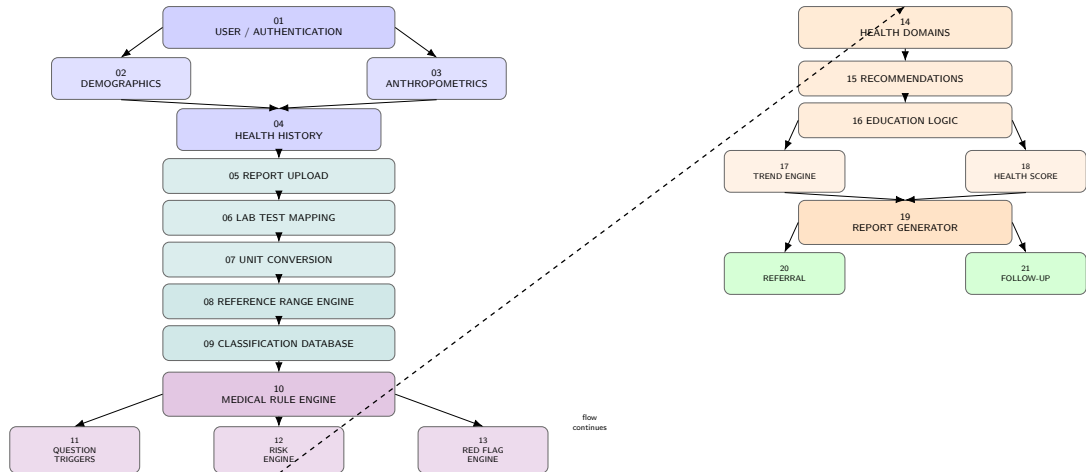
Modular, not screen-based

- Do not arrange the platform purely by screen
- Arrange into separate modules so UI, medical logic, laboratory data, risk algorithms, and reporting can each evolve independently
- This document defines the master technical module architecture: 26 modules, grouped into 6 functional clusters



Master Technical Architecture (Modules 01–21)

The full pipeline, top to bottom



System-Wide Governance Layer (Modules 22–26)

Five modules underlying the entire pipeline

Supports every module above (01–21) – versioning, audit, security and analytics run underneath the entire pipeline

22
Medical Admin /
Knowledge Mgmt

23
Version
Control

24
Audit / Decision
Log

25
Security / Privacy /
Permissions

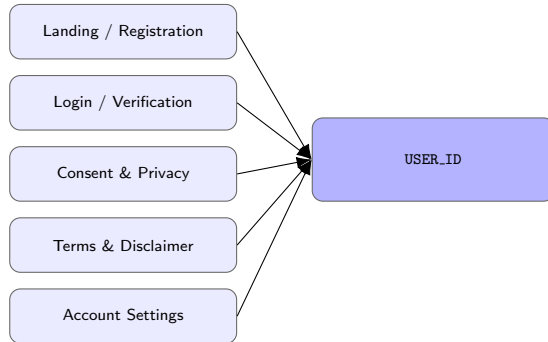
26
Analytics

Module 1: User Access & Welcome Module

Cluster 1 – Identity & Profile (M1–M4)

Every intake screen resolves into one identity

- Entry layer: landing, registration, login, verification, consent, privacy, terms, disclaimer, account settings, language
- Main output: USER_ID
- Almost everything afterward links back to this one ID

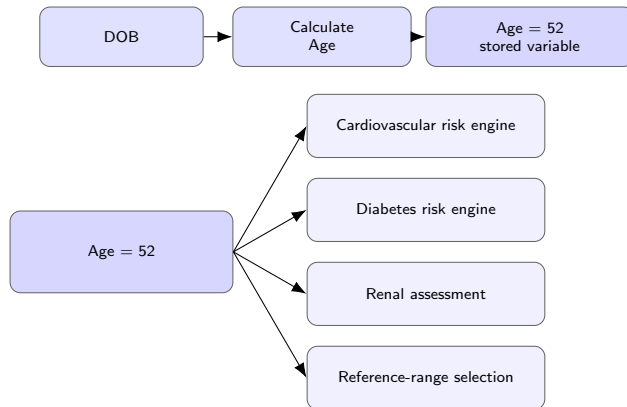


Module 2: Basic Demographic Profile Module

Cluster 1 – **Identity & Profile** (M1–M4)

One demographic value, reused by four downstream engines

- Stores DOB, calculated age, biological sex, ethnicity (where relevant), region, pregnancy/menopausal status
- Becomes part of the shared risk-variable database

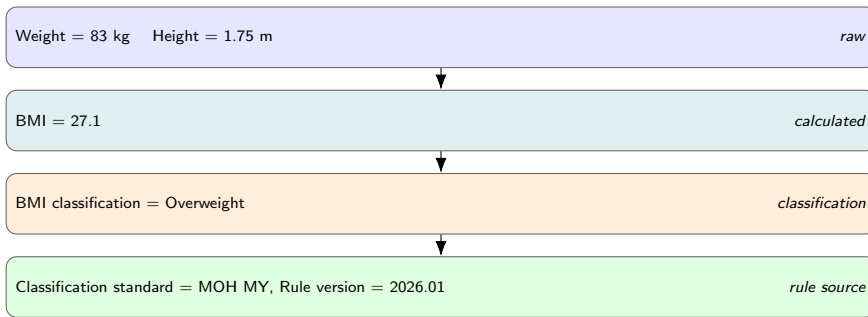


Module 3: Anthropometric Profile Module

Cluster 1 – Identity & Profile (M1–M4)

Four layers, stored separately, so guideline updates never touch raw data

- Own module because it produces longitudinal anthropometric trends
- Raw: height, weight, waist/hip circumference, BP, heart rate, body composition
- Derived: BMI, BMI category, waist-height/waist-hip ratio, central obesity, BP classification, weight change %
- **Key rule: raw data and calculated classifications are stored separately**

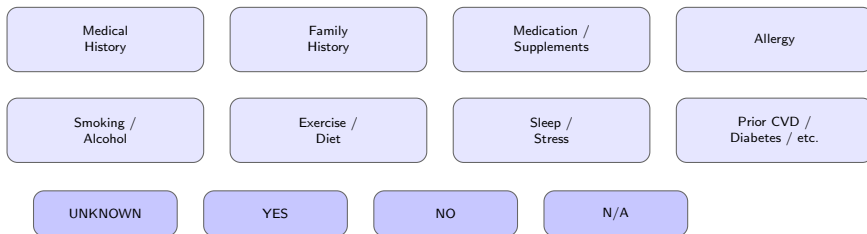


Module 4: Health History Module

Cluster 1 – Identity & Profile (M1–M4)

Progressive categories, each answerable in four defined states

- Progressive collection, not all at registration: medical/family history, medication, supplements, allergy, smoking, alcohol, exercise, diet, sleep, stress, prior CVD/diabetes/hypertension/kidney/liver disease
- **Answer states: UNKNOWN / YES / NO / NOT APPLICABLE** – never let NULL mean everything

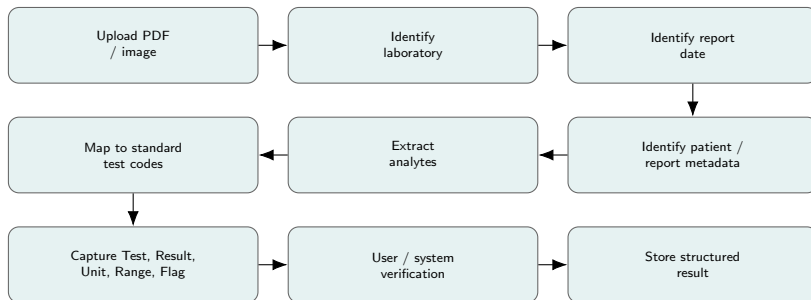


Module 5: Blood Report Upload Module

Cluster 2 – Data Ingestion & Standardisation (M5–M9)

Nine-step ingestion pipeline

- The ingestion layer: upload → identify lab/date/metadata → extract analytes → map to standardized codes → capture test/result/unit/range/flag → verify → store
- Design the database as though every result eventually becomes structured data, even before OCR is fully automated



Module 5: Blood Report Upload Module

One fully structured result

Field	Value
Test	ALT
Value	68
Unit	U/L
Lab lower limit	0
Lab upper limit	41
Lab	XYZ Laboratory
Date	2026-08-20

Module 6: Laboratory Test Master Database

Cluster 2 – **Data Ingestion & Standardisation** (M5–M9)

Standardized dictionary example

- Standardized Lab Test Dictionary solves inconsistent lab naming conventions
- Downstream algorithms reference the internal code (e.g. LAB_HBA1C) – never the raw report text

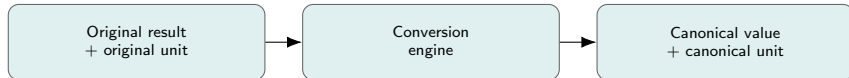
Internal Code	Standard Name	Alternate Names
LAB_GLU_FAST	Fasting Plasma Glucose	FBS, FPG
LAB_HBA1C	HbA1c	Glycated Hb
LAB_ALT	ALT	SGPT
LAB_AST	AST	SGOT
LAB_CREAT	Creatinine	Serum Creatinine

Module 7: Unit Standardisation & Conversion Module

Cluster 2 – Data Ingestion & Standardisation (M5–M9)

Glucose example – both values are kept

- Algorithms must never consume whatever unit the lab happens to report
- Original result + original unit is preserved; a canonical value + canonical unit is derived alongside it
- **Never overwrite the original laboratory value**



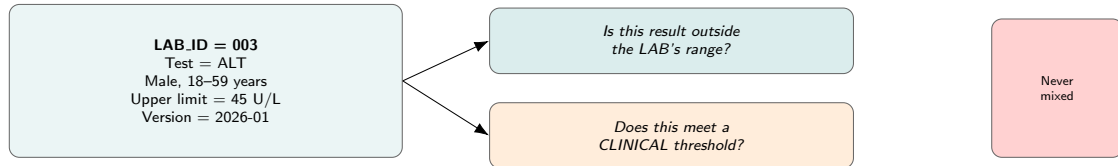
Original: Glucose = 108 mg/dL → Canonical: Glucose = 6.0 mmol/L (both kept)

Module 8: Laboratory Reference Range Database

Cluster 2 – Data Ingestion & Standardisation (M5–M9)

Two distinct questions, two distinct databases

- Separate from the disease/risk classification database
- Stores lab, test, limits, unit, sex/age/pregnancy criteria, method, effective date, version
- Answers “is this outside the *lab*’s range?” – a different question from “does this meet a *clinical* threshold?”



Module 9: Clinical Classification Standard Database

Cluster 2 – Data Ingestion & Standardisation (M5–M9)

One classification rule, fully specified as metadata

- Where medical guidelines are stored: BMI, BP, HbA1c, fasting glucose, LDL, triglyceride, eGFR, albuminuria, anaemia, liver enzyme classifications, etc.
- Every rule carries full metadata – this becomes the medical knowledge database

RULE_ID: GLYCEMIA_HBA1C_001

Parameter: HbA1c

Condition: <5.7%

Classification: Normal

Guideline: Specified clinical guideline

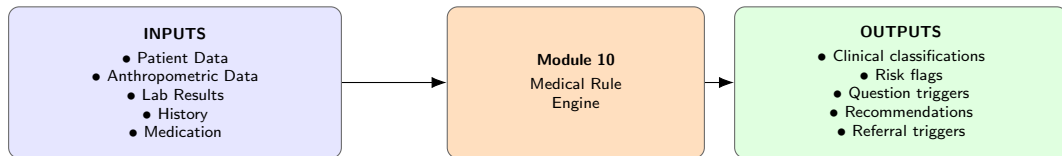
Version: 2026.01

Module 10: Rule Engine / Medical Logic Engine

Cluster 3 – **Medical Intelligence Core** (M10–M13)

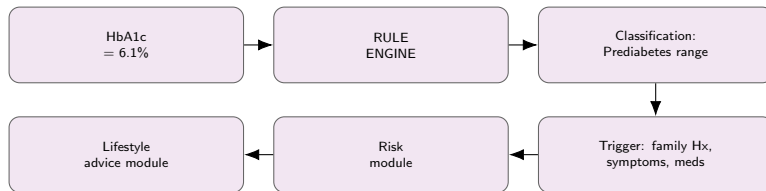
Five inputs, five outputs, one engine

- **The heart of the platform** – medical decisions must never be coded directly into individual screens
- Combines patient, anthropometric, lab, history and medication data into classifications, risk flags, question triggers, recommendations and referral triggers



Module 10: Rule Engine / Medical Logic Engine

One value, traced through the engine

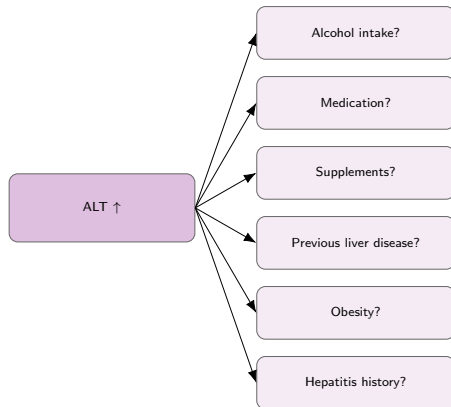


Module 11: Question Trigger Engine

Cluster 3 – **Medical Intelligence Core** (M10–M13)

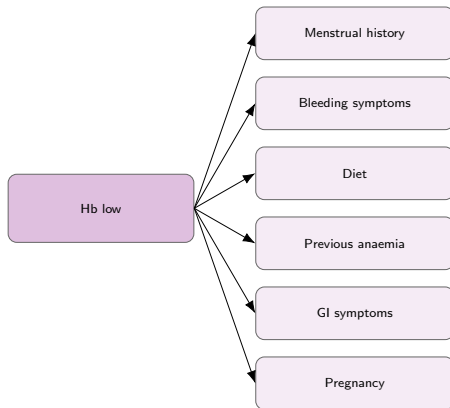
Example 1 – ALT elevated

- Independent module – users should not answer 80 questions at registration
- An abnormal finding activates only the relevant question set
- This is the module that makes the platform feel “intelligent”



Module 11: Question Trigger Engine

A second finding, a different question set

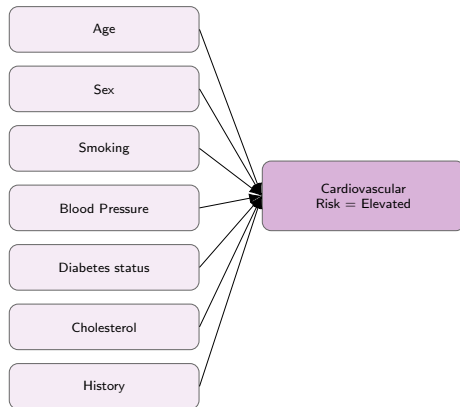


Module 12: Risk Stratification Engine

Cluster 3 – Medical Intelligence Core (M10–M13)

Many classification inputs converge into one risk output

- Classification and risk must stay separate: “LDL = high” is a classification, not a risk score
- Age + sex + smoking + BP + diabetes + cholesterol + history together may produce “Cardiovascular risk = elevated”
- Each calculator needs an ID, required variables, formula, exclusions, interpretation, version, source guideline

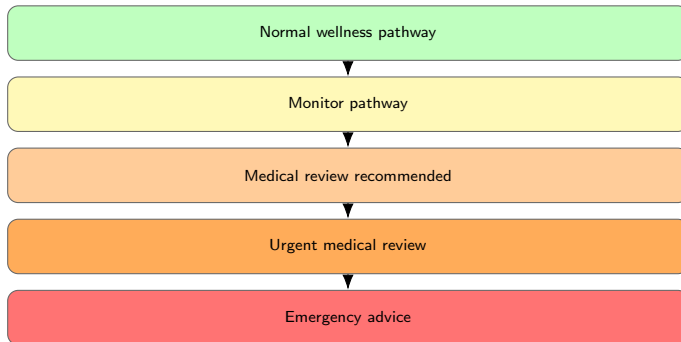


Module 13: Red Flag / Escalation Module

Cluster 3 – **Medical Intelligence Core** (M10–M13)

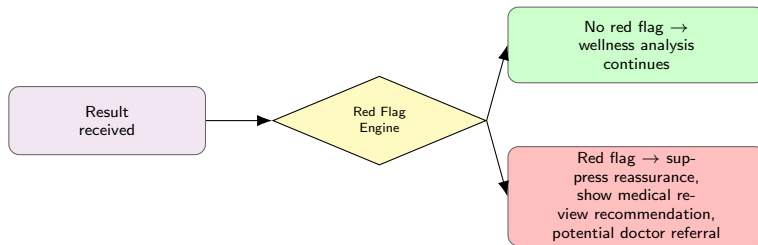
Five-stage escalation ladder

- Separates Normal wellness → Monitor → Medical review → Urgent review → Emergency advice
- Red flags are centrally controlled and versioned, never scattered across screens



Module 13: Red Flag / Escalation Module

One gate, two possible outcomes

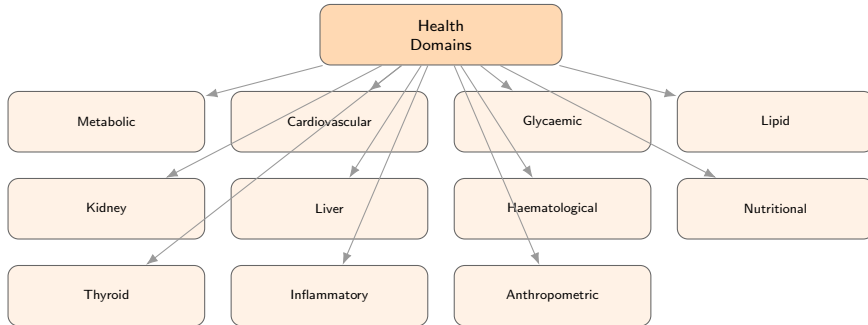


Module 14: Health Domain Analysis Module

Cluster 4 – Interpretation & Reporting (M14–M19)

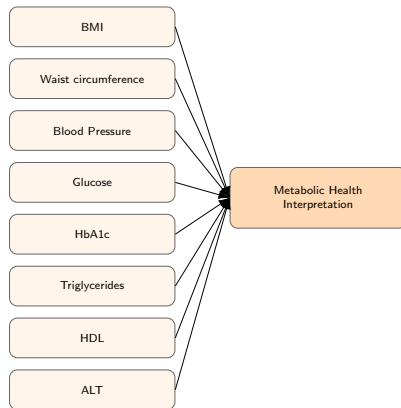
Eleven interpretable health domains

- Combines ~40 individual tests into interpretable domains rather than a raw list
- Metabolic, cardiovascular, glycaemic, lipid, kidney, liver, haematological, nutritional, thyroid, inflammatory, anthropometric health
- Makes the report far easier for patients to understand



Module 14: Health Domain Analysis Module

Eight parameters, one domain interpretation

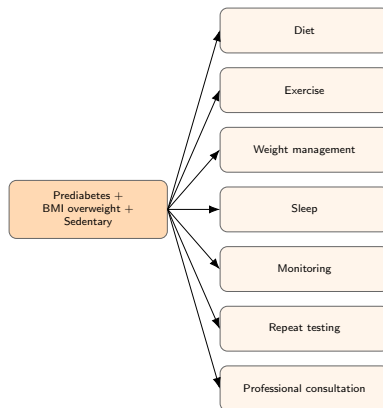


Module 15: Recommendation Engine

Cluster 4 – Interpretation & Reporting (M14–M19)

One finding combination, seven candidate recommendations

- Recommendations are separate from classifications – one classification can trigger several recommendations
- Recommendations use codes (e.g. REC_DIET_001) instead of hard-coded text

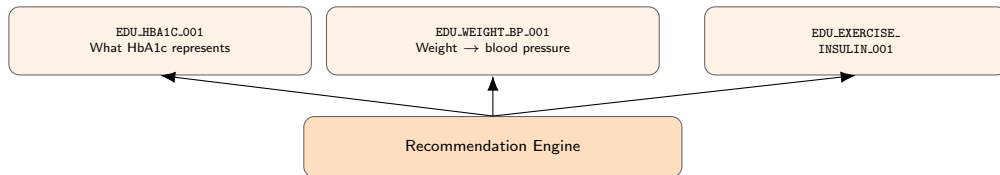


Module 16: Explanation / Health Education Library

Cluster 4 – **Interpretation & Reporting** (M14–M19)

Recommendation Engine calls the education library

- A dedicated content library so the platform can explain *why* advice is given
- The Recommendation Engine calls these entries – doctors can update wording without touching code

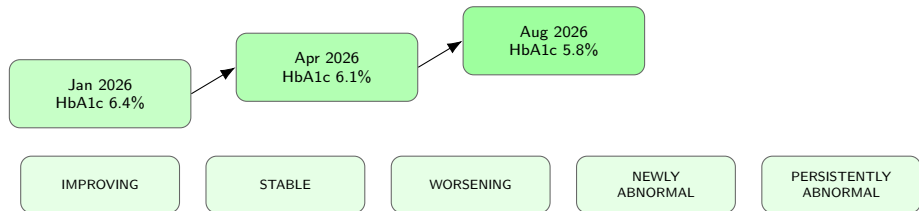


Module 17: Trend & Longitudinal Health Module

Cluster 4 – **Interpretation & Reporting** (M14–M19)

A falling HbA1c series classified as Improving

- Every parameter is stored longitudinally, not just archived as a PDF
- Computes absolute/percentage change, direction, rate of change, persistent/new/resolved abnormality

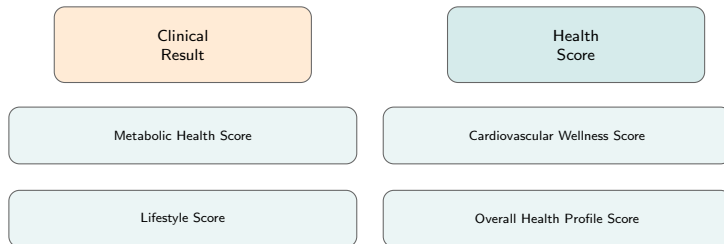


Module 18: Health Score Module

Cluster 4 – Interpretation & Reporting (M14–M19)

A communication layer, never a substitute for classification

- If used, kept separate from clinical classifications: metabolic, cardiovascular wellness, lifestyle, overall profile scores
- **Clinical result** $\neq$ **health score** – the score communicates, it does not replace medical classification

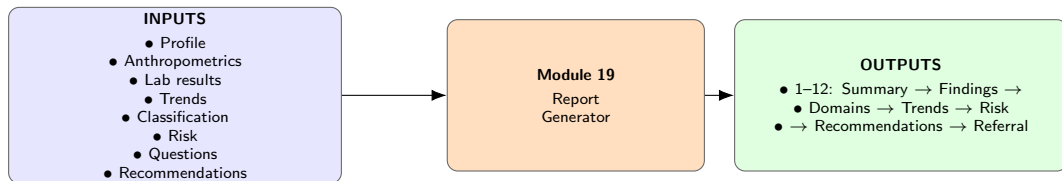


Module 19: Report Generation Module

Cluster 4 – **Interpretation & Reporting** (M14–M19)

Eight inputs assembled into a twelve-section report

- Assembles every other module's output into one report
- Twelve-section structure: summary, findings, anthropometrics, results, domains, trends, risk factors, attention areas, recommendations, rationale, monitoring, referral

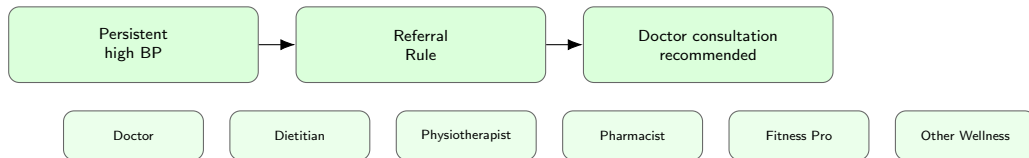


Module 20: Referral & Professional Network Module

Cluster 5 – **Engagement** (M20–M21)

Rule-triggered referral, not advertising

- Later supports doctor, dietitian, physiotherapist, pharmacist, fitness professional, other wellness professionals
- Triggered by medical logic, not arbitrary advertising

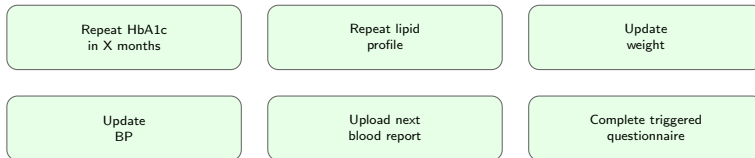


Module 21: Notification & Follow-Up Module

Cluster 5 – **Engagement** (M20–M21)

Six example follow-up triggers

- Creates longitudinal engagement instead of a one-time blood-report service

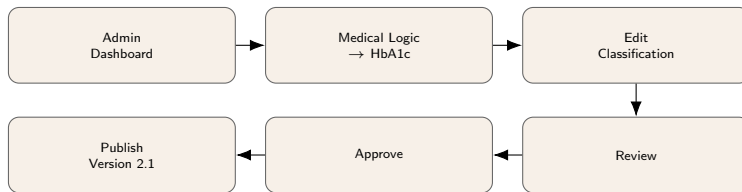


Module 22: Medical Knowledge Management / Admin Module

Cluster 6 – Platform Governance (M22–M26)

Medical content changes without a code redeploy

- An interface for the medical team to manage reference ranges, classifications, risk algorithms, red flags, triggers, recommendations, education content, guideline versions
- Programmers should not need to redeploy the whole application when a medical threshold changes



Module 23: Version Control Module

Cluster 6 – **Platform Governance** (M22–M26)

Every report carries its own version stamp

- Critical for medical software: every report records the medical rule, reference-range, risk-calculator, recommendation and report-engine versions it used
- Lets the team determine exactly why an old report produced its conclusion after a guideline changes

Report generated: 24 Aug 2026

Medical Knowledge Base: v1.4

Cardiovascular Algorithm: v2.1

Cluster 6 – Platform Governance (M22–M26)

A full decision trace, retained per conclusion

- Every important conclusion retains its reasoning inputs
- Valuable for debugging, medical review, quality assurance, dispute investigation, and algorithm improvement

USER 123

HbA1c = 6.1 Rule triggered = GLU_004

Classification = Prediabetes range

Trigger questions = Q_GLU_01, Q_GLU_04

Recommendation = REC_GLU_02 Rule version = 1.3

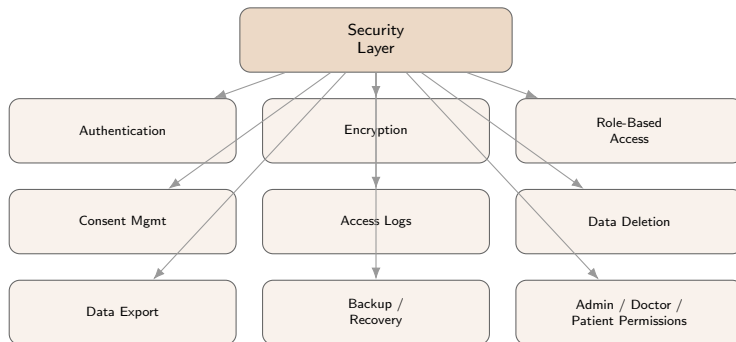
Timestamp = ...

Module 25: Security, Privacy & Permission Module

Cluster 6 – Platform Governance (M22–M26)

Nine components under one platform-wide layer

- A platform-wide backend layer: authentication, encryption, role-based access, consent management, access logs, deletion, export, backup, recovery
- Medical administrators, programmers and customer support each need different permission levels

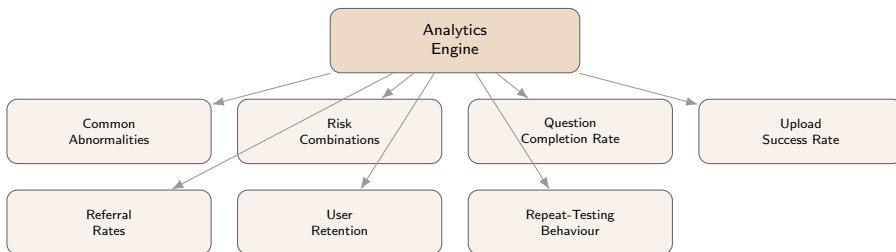


Module 26: Analytics & System Improvement Module

Cluster 6 – Platform Governance (M22–M26)

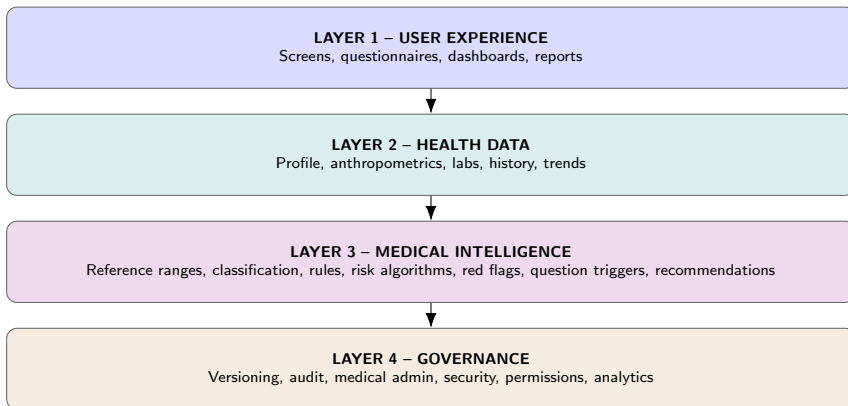
System-level insight, never clinical decision-making

- Anonymised, system-level analytics: common abnormalities, risk combinations, question completion rate, upload success rate, referral rates, retention, repeat-testing behaviour
- Kept separate from clinical decision-making



The Four-Layer Model

Think of the platform as four layers, not one application



Never Hard-Code Medical Knowledge

The single most important rule for developers

DO NOT

```
if hba1c >= 6.5:  
    diagnosis = 'diabetes'  
  
...repeated throughout the application
```

INSTEAD

HbA1c → Medical Rule Engine → Retrieve
active rule → Apply criteria → Return
classification code → UI displays approved wording

Scalable: update WHO/MOH guideline standards, introduce new risk calculators, add new blood tests, or change recommendation wording – without rebuilding the platform.

Team Division (Four-Person Team)

This architecture creates a clean ownership split

Medical Logic Person

Owns Modules 6–16
+ medical content of
Modules 22–24

Technical Developers

Build the engines,
databases, APIs and UI
(all 26 modules)

UI / Marketing Person

Designs Modules 1–5
+ presentation of
Modules 14–21

Thank You

MyHealthReport — Technical Module Architecture